



Chulalongkorn University

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template from KACTL

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Template (1)

```
template.cpp31 lines

#pragma once
#include <bits/stdc++.h>
#define sz(x) (int)(x).size()
#define all(x) (x).begin(), (x).end()

using namespace std;

typedef long long ll;
typedef double db;
typedef long double ld;
typedef pair<int, int> pii;
typedef pair<ll, ll> pll;

template<typename T> bool ckmin(T &a, const T &b) { return b < a ? a = b, 1 : 0; }
template<typename T> bool ckmax(T &a, const T &b) { return a < b ? a = b, 1 : 0; }

mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());

const char nl = '\n';
const int INF = 0x3fffffff;
const int MOD=1000000007;
// const int MOD = 998244353;
const ll LINF = 0xffffffffffffffff;
const db DINF = numeric_limits<db>::infinity();
const db EPS = 1e-9;
const db PI = acos(db(-1));
```

```
1signed main(){
1    ios_base::sync_with_stdio(0); cin.tie(NULL);
1    return 0;
1}

c.sh2 lines

g++ -std=gnu++2a -Wall $1 -o a.out
./a.out
```

Mathematics (2)

2.1 Goldbatch’s Conjecture

- Even number can be written in sum of two primes (Up to 1e12)
- Range of N^{th} prime and $N + 1^{th}$ prime will be less than or equal to 300 (Up to 1e12)

2.2 Divisibility

Number of divisors of N is given by $\prod_{i=1}^k (a_i + 1)$ where $N = \prod_{i=1}^k p_i^{a_i}$ and p_i are prime factors of N .

Combinatorial (3)

3.1 Permutations

3.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800
n	11	12	13	14	15	16	17			
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14			
n	20	25	30	40	50	100	150	171		
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

IntPerm.h 3.1.2 Cycles

Let $g_S(n)$ be the number of n -permutations whose cycle lengths all belong to the set S . Then

$$\sum_{n=0}^{\infty} g_S(n) \frac{x^n}{n!} = \exp \left(\sum_{n \in S} \frac{x^n}{n} \right)$$

3.1.3 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1) + D(n-2)) = nD(n-1) + (-1)^n = \left\lfloor \frac{n!}{e} \right\rfloor$$

3.1.4 Burnside’s lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k|n} f(k) \phi(n/k).$$

3.2 Partitions and subsets

3.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	$\sim 2e5$	$\sim 2e8$

3.2.2 Lucas’ Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + \dots + n_1 p + n_0$ and $m = m_k p^k + \dots + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod{p}$.

3.2.3 Binomials

multinomial.h

3.3 General purpose numbers

3.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able).
 $B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$

Sums of powers:

$$\sum_{i=1}^n n^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\begin{aligned} \sum_{i=m}^{\infty} f(i) &= \int_m^{\infty} f(x) dx - \sum_{k=1}^{\infty} \frac{B_k}{k!} f^{(k-1)}(m) \\ &\approx \int_m^{\infty} f(x) dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m)) \end{aligned}$$

3.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n,k)=c(n-1,k-1)+(n-1)c(n-1,k),\; c(0,0)=1$$
$$\sum_{k=0}^nc(n,k)x^k=x(x+1)\ldots(x+n-1)$$

$$c(8,k)=8,0,5040,13068,13132,6769,1960,322,28,1$$
$$c(n,2)=0,0,1,3,11,50,274,1764,13068,109584,\ldots$$

3.3.3 Eulerian numbers

Number of permutations $\pi\in S_n$ in which exactly k elements are greater than the previous element. $k\;j$:s s.t. $\pi(j)>\pi(j+1)$, $k+1\;j$:s s.t. $\pi(j)\geq j$, $k\;j$:s s.t. $\pi(j)>j$.

$$E(n,k)=(n-k)E(n-1,k-1)+(k+1)E(n-1,k)$$
$$E(n,0)=E(n,n-1)=1$$
$$E(n,k)=\sum_{j=0}^k(-1)^j\binom{n+1}{j}(k+1-j)^n$$

3.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n,k)=S(n-1,k-1)+kS(n-1,k)$$
$$S(n,1)=S(n,n)=1$$
$$S(n,k)=\frac{1}{k!}\sum_{j=0}^k(-1)^{k-j}\binom{k}{j}j^n$$

3.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n)=1,1,2,5,15,52,203,877,4140,21147,\ldots$. For p prime,

$$B(p^m+n)\equiv mB(n)+B(n+1)\pmod{p}$$

3.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i : $n_1n_2\cdots n_kn^{k-2}$
with degrees d_i : $(n-2)!/((d_1-1)!\cdots(d_n-1)!)$

3.3.7 Catalan numbers

$$C_n=\frac{1}{n+1}\binom{2n}{n}=\binom{2n}{n}-\binom{2n}{n+1}=\frac{(2n)!}{(n+1)!n!}$$
$$C_0=1,\; C_{n+1}=\frac{2(2n+1)}{n+2}C_n,\; C_{n+1}=\sum C_iC_{n-i}$$
$$C_n=1,1,2,5,14,42,132,429,1430,4862,16796,58786,\ldots$$

- sub-diagonal monotone paths in an $n\times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n+1$ leaves (0 or 2 children).

- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Data Structures (4)

OrderedSet.hpp

SparseTable.hpp

FenwickTree.hpp

2DFenwickTree.hpp

SegmentTree.hpp

LazySegmentTree.hpp

DynamicSegmentTree.hpp

PersistentSegmentTree.hpp

LiChaoTree.hpp

BinaryTrie.hpp

Mo.hpp

StaticTopTree.hpp

Treap.hpp

Number Theory (5)

ExtendedEuclid.hpp

euclid.h

CRT.hpp

phiFunction.hpp

FloorSum.hpp

5.1 Prime Numbers

MillerRabin.hpp

LinearSieve.hpp

FastEratosthenes.hpp

GolbatchConjecture.hpp

5.2 Modulo

ModArith.hpp

ModGen.hpp

ModInverse.hpp

ModLog.hpp

ModMulLL.hpp

ModPow.hpp

ModSqrt.hpp

ModSum.hpp

ModArithmetic.hpp

Graph (6)

SCC.hpp

ArticulationPoint.hpp

Bridge.hpp

Hierholzer.hpp

6.1 Matching

HopcroftKarp.hpp

Kuhn.hpp

WeightedMatching.hpp

6.2 Network Flow

Dinic.hpp

MinCostFlow.hpp

Tree (7)

LCA.hpp

CartesianTree.hpp

VirtualTree.hpp

HLD.hpp

CentroidDecom.hpp

RootedTreeIsomorphism.hpp

UnrootedTreeIsomorphism.hpp

Strings (8)

KMP.hpp

ZAlgo.hpp

AhoCorasick.hpp

Ukkonen.hpp

SuffixArray.hpp

PrefixFunction.hpp

SuffixAutomaton.hpp

Geometry (9)

9.1 Geometric primitives

Point.h

lineDistance.h

SegmentDistance.h

SegmentIntersection.h

lineIntersection.h

sideOf.h

OnSegment.h

linearTransformation.h

LineProjectionReflection.h

Angle.h

9.2 Circles

CircleIntersection.h

CircleTangents.h

CircleLine.h

CirclePolygonIntersection.h

circumcircle.h

MinimumEnclosingCircle.h

9.3 Polygons

InsidePolygon.h

PolygonArea.h

PolygonCenter.h

PolygonCut.h

ConvexHull.h

HullDiameter.h

PointInsideHull.h

LineHullIntersection.h

9.4 Misc. Point Set Problems

ClosestPair.h

ManhattanMST.h

kdTree.h

DelaunayTriangulation.h

FastDelaunay.h

9.5 3D

PolyhedronVolume.h

Point3D.h

3dHull.h

sphericalDistance.h

ComplexGeometry.hpp

DefiniteIntegral.hpp

Dynamic Programming (10)

CHT.hpp

Knuth.hpp

DVC.hpp

SlopeTrick.hpp

AlienTrick.hpp

Polynomials (11)

11.1 Newton’s Method

if $F(Q) = 0$, then $Q_{2n} \equiv Q_n - \frac{F(Q_n)}{F'(Q_n)} \pmod{x^{2n}}$

$$Q = P^{-1} : Q_{2n} \equiv Q_n \cdot (2 - P \cdot Q_n^2) \pmod{x^{2n}}$$

$$Q = \ln P = \int \frac{P'}{P} dx$$

$$Q = e^P : Q_{2n} \equiv Q_n(1 + P - \ln Q_n) \pmod{x^{2n}}$$

$$Q = \sqrt{P} : Q_{2n} \equiv \frac{1}{2}(Q_n + P \cdot Q_n^{-1}) \pmod{x^{2n}}$$

$$Q = P^k = \alpha^k x^{kt} e^{k \ln T}; P = \alpha \cdot x^t \cdot T, T(0) = 1$$

Interpolation.hpp

FormalPowerSeries.hpp

FFTTNTT.hpp

Convolutions (12)

AndConvolution.hpp

GCDConvolution.hpp

LCMConvolution.hpp

ORConvolution.hpp

XORConvolution.hpp

MaxPlusConvolution.hpp

Various (13)

GaussianElimination.hpp

XORBasis.hpp

RangeXor.hpp

SOS.hpp

13.1 Optimization tricks

`__builtin_ia32_ldmxcsr(40896);` disables denormals (which make floats 20x slower near their minimum value).

13.1.1 Bit hacks

- `x & -x` is the least bit in `x`.
- `for (int x = m; x; ) { --x &= m; ... }` loops over all subset masks of `m` (except `m` itself).
- `c = x&-x, r = x+c; (((r^x) >> 2)/c) | r` is the next number after `x` with the same number of bits set.
- `rep(b,0,K) rep(i,0,(1 << K)) if (i & 1 << b) D[i] += D[i^(1 << b)];` computes all sums of subsets.

13.1.2 Pragmas

- `#pragma GCC optimize ("Ofast")` will make GCC auto-vectorize loops and optimizes floating points better.
- `#pragma GCC target ("avx2")` can double performance of vectorized code, but causes crashes on old machines.
- `#pragma GCC optimize ("trapv")` kills the program on integer overflows (but is really slow).